

Homework #6

[Your name]

[Total - 100 pts]

Question 1: For each of the following, (i) verify each distribution belongs to the exponential family, (ii) find the MLE for θ , and (iii) find the limiting distribution of the MLE.

- (a) Beta distribution: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Beta}(\theta, 1)$, $\theta > 0$. [Hint: note that $-\theta \log(X) \sim \text{Exp}(1)$ if $X \sim \text{Beta}(\theta, 1)$]
- (b) Pareto distribution: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Pareto}(1, \theta)$, $\theta > 2$. [Hint: note that $\theta \log(X) \sim \text{Exp}(1)$ if $X \sim \text{Pareto}(1, \theta)$]

[1.a] Solution

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[1.b] Solution

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Question 2: Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Uniform}(-\theta, \theta)$ where $\theta > 0$. Answer the following:

- (a) Show that the statistic $Y = \max\{|X_i|\}$ for $1 \leq i \leq n$ and $\theta \in (0, +\infty)$ is a sufficient statistic.
- (b) Denote the order statistics as $X_{(1)} < \dots < X_{(n)}$ for $n \geq 2$, and define an estimator for θ as:

$$\hat{\theta} = c_n \cdot (X_{(n)} - X_{(1)})$$

Find the value of c_n such that the estimator is unbiased for θ (i.e., $\mathbb{E}(\hat{\theta}) = \theta$).

[2.a] Solution

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[2.b] Solution

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Question 3: Find a sufficient statistic:

- (a) $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\theta)$, for $\theta > 0$, where the pdf is given by

$$\text{pdf}(x; \theta) = \frac{1}{\theta} e^{-x/\theta} \cdot \mathbf{1}\{x \in (0, \infty)\}.$$

(b) $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$, for $\alpha > 0$ and $\beta > 0$, where the pdf is given by

$$\text{pdf}(x; \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} \cdot \mathbf{1}\{x \in (0, \infty)\}.$$

[3.a] Solution

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[3.b] Solution

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Question 4: Consider an exponential distribution $\text{Exp}(\mu, \sigma)$ with two parameters, where $-\infty < \mu < +\infty$ and $\sigma > 0$. The marginal pdf is given by

$$\text{pdf}(x; \mu, \sigma) = \frac{1}{\sigma} e^{-(x-\mu)/\sigma} \cdot \mathbf{1}\{x \in (\mu, \infty)\}.$$

Suppose that you are unable to observe the individual random samples, but you can observe their order statistics $X_{(1)} < \dots < X_{(r)}$ for $1 \leq r < n$.

(a) Does $\text{Exp}(\mu, \sigma)$ belong to the exponential family? Explain.

(b) Find a sufficient statistic for (μ, σ) , where $(\mu, \sigma) \in \mathbb{R} \times \mathbb{R}_+$.

Hint: the joint density of the first r order statistics (say $Y_1 < Y_2 < \dots < Y_r$) when $r < n$ is given by

$$f_{Y_1, \dots, Y_r}(y_1, \dots, y_r) = \frac{n!}{(n-r)!} \left[\prod_{i=1}^r f_X(y_i) \right] [1 - F_X(y_r)]^{n-r}$$

[4.a] Solution

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[4.b] Solution

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